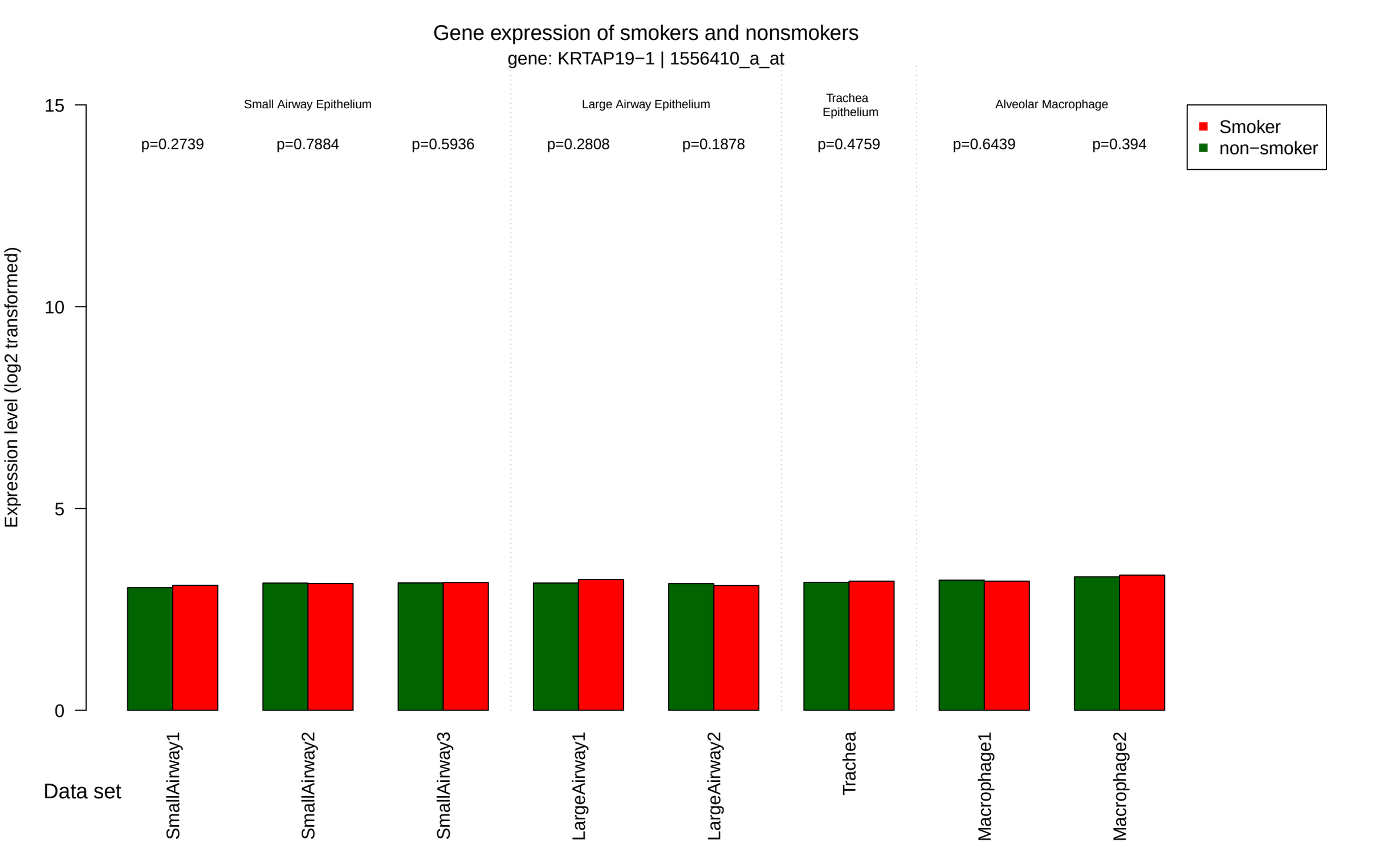
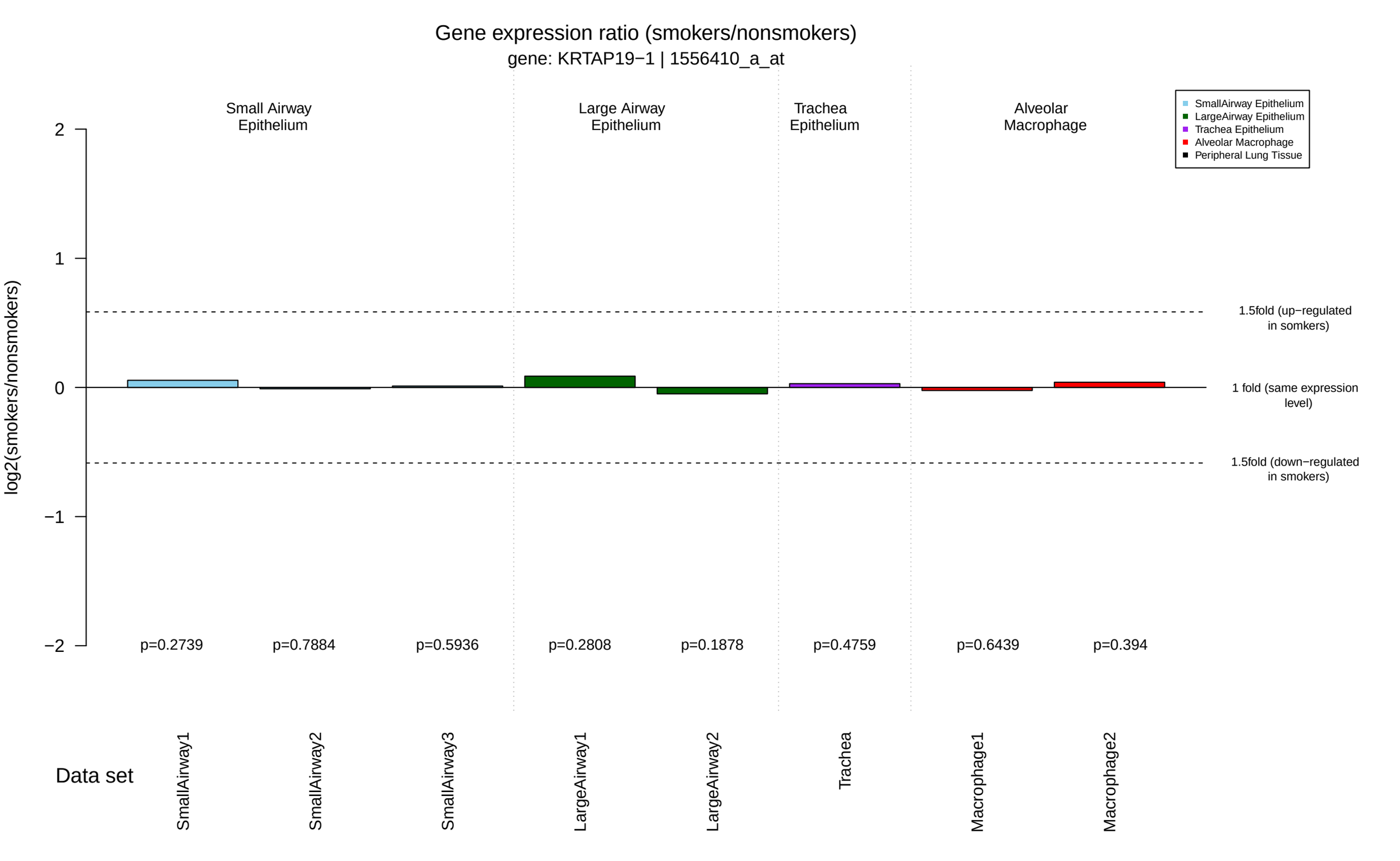
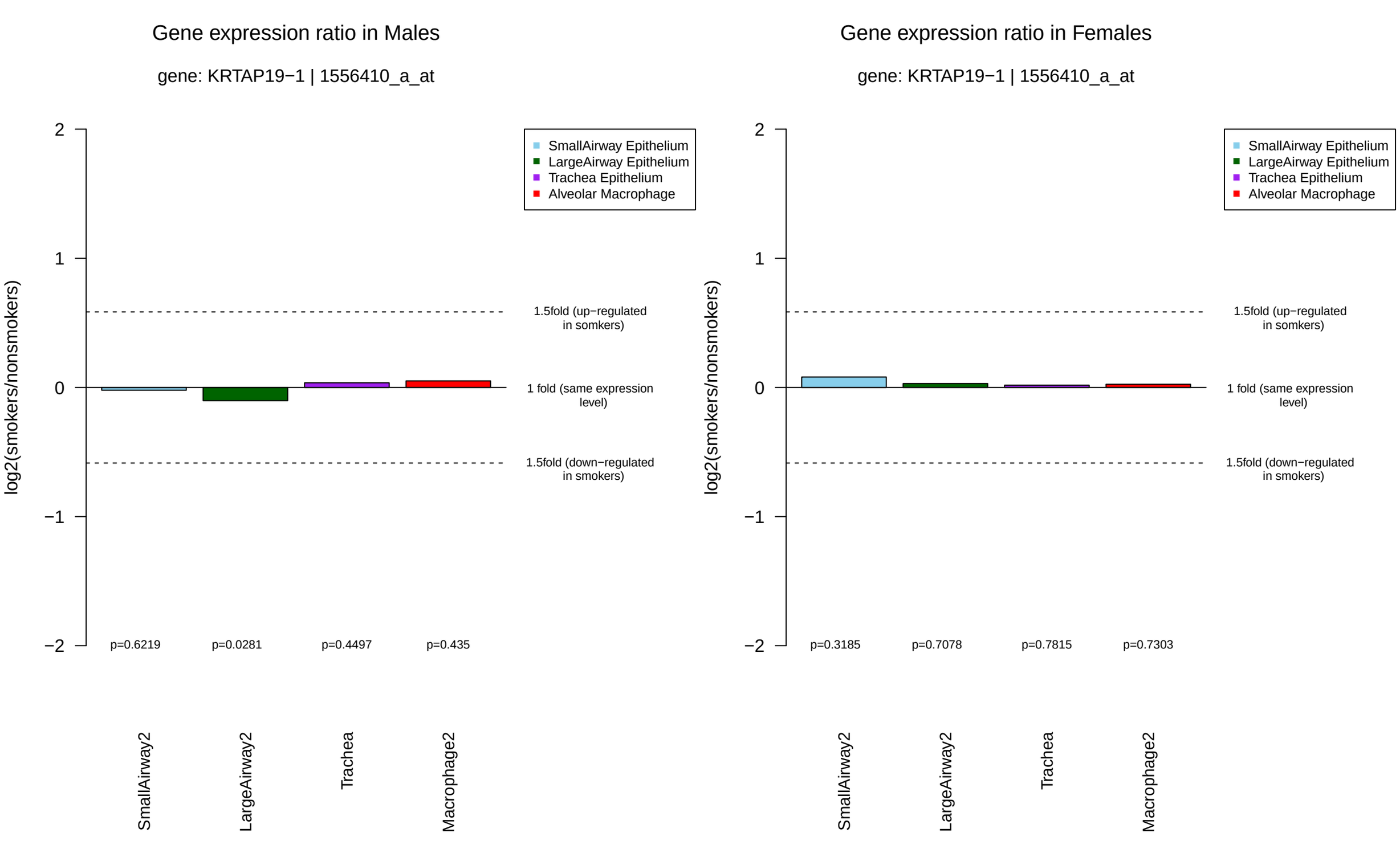
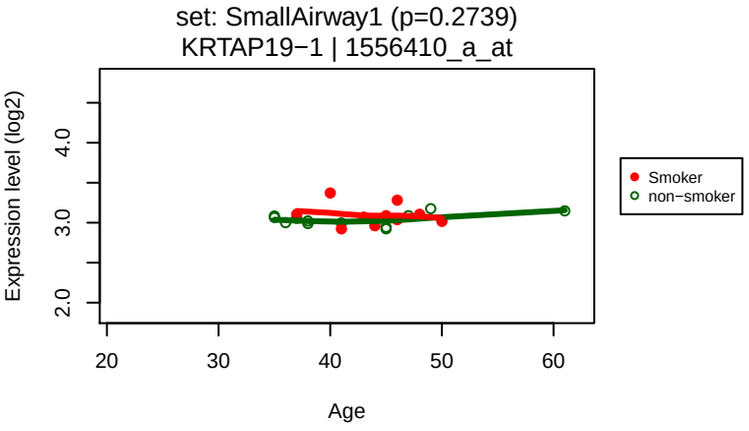
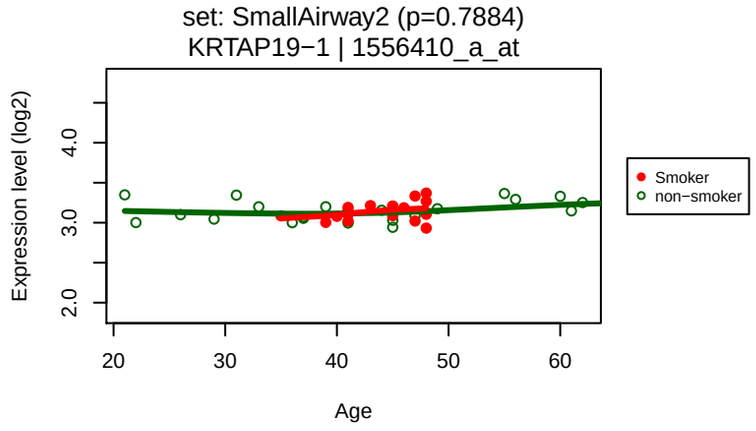
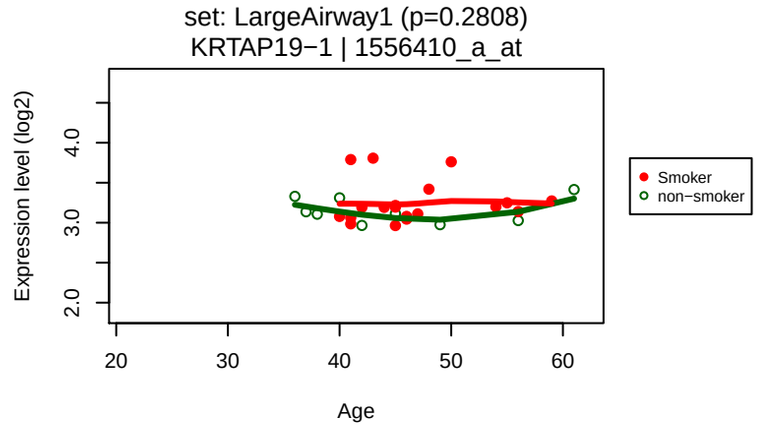

Gene expression of smokers and nonsmokers
gene: KRTAP19−1 | 1556410_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2739
p=0.7884
p=0.5936
p=0.2808
p=0.1878
p=0.4759
p=0.6439
p=0.394
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

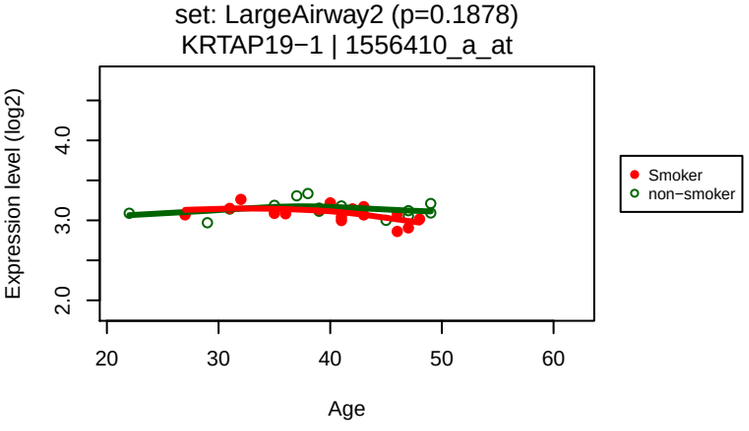

Gene expression ratio (smokers/nonsmokers)
gene: KRTAP19−1 | 1556410_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2739
p=0.7884
p=0.5936
p=0.2808
p=0.1878
p=0.4759
p=0.6439
p=0.394
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

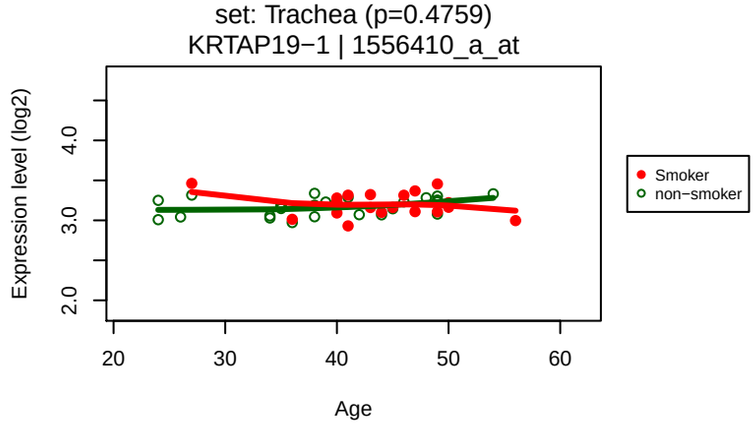

Gene expression ratio in Males
gene: KRTAP19−1 | 1556410_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6219
p=0.0281
p=0.4497
p=0.435
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRTAP19−1 | 1556410_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3185
p=0.7078
p=0.7815
p=0.7303
SmallAirway2
LargeAirway2
Trachea
Macrophage2

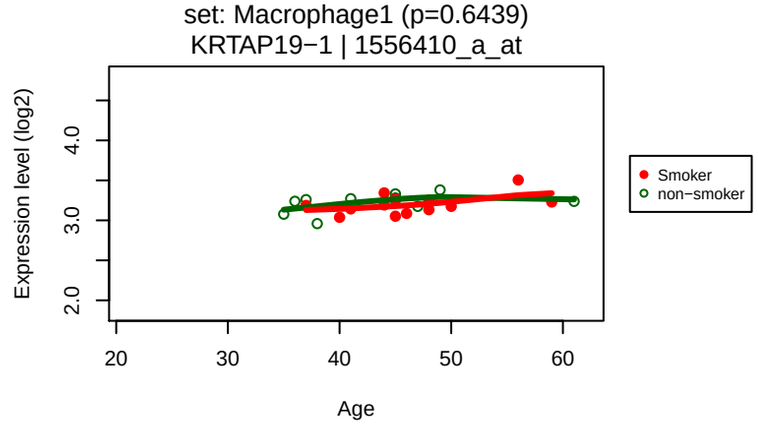

set: SmallAirway1 (p=0.2739)
KRTAP19−1 | 1556410_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

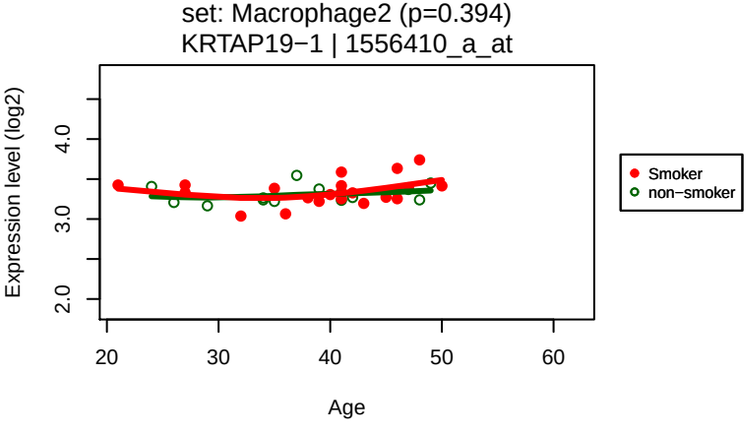

set: SmallAirway2 (p=0.7884)
KRTAP19−1 | 1556410_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

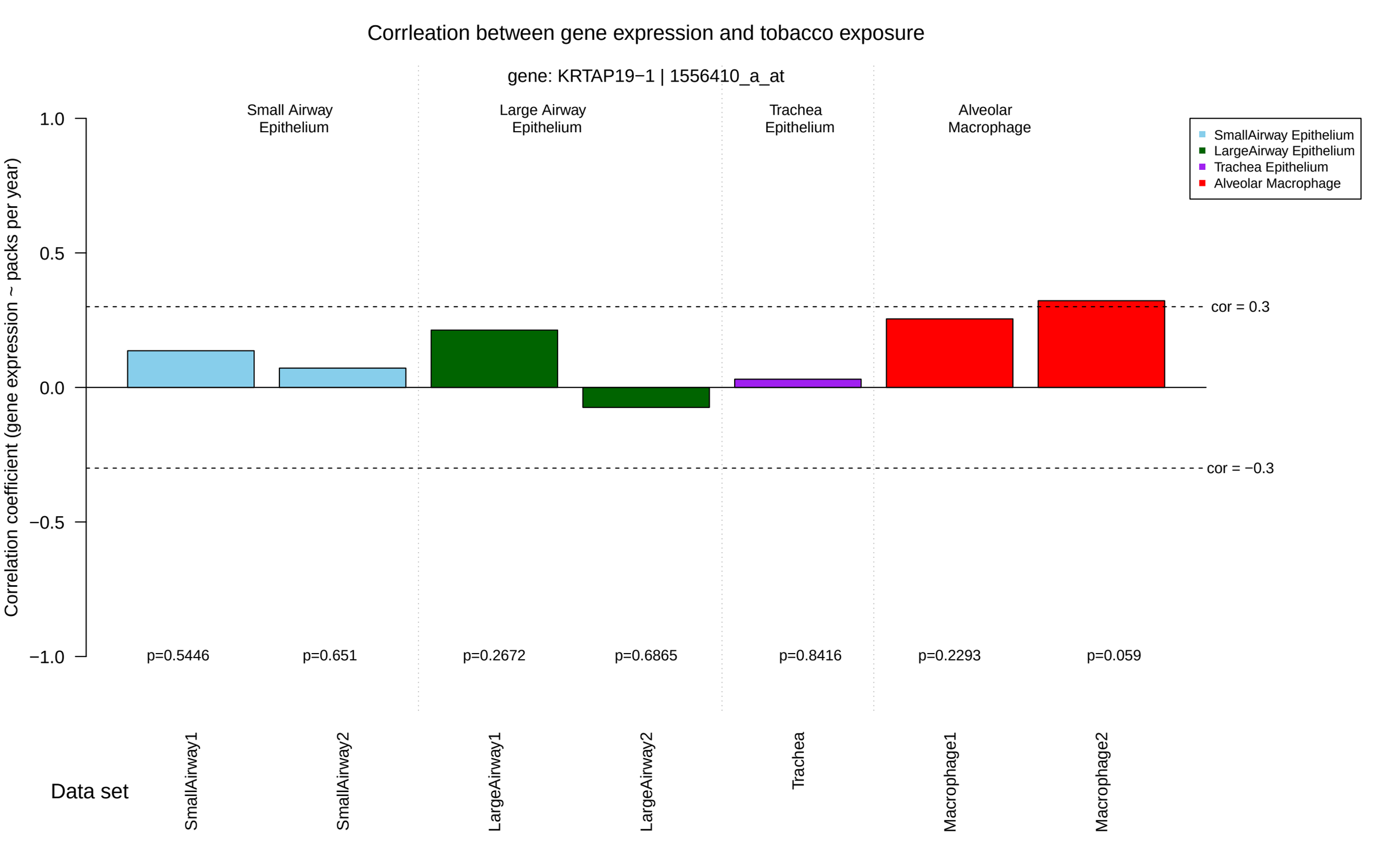

set: LargeAirway1 (p=0.2808)
KRTAP19−1 | 1556410_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1878)
KRTAP19−1 | 1556410_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.4759)
KRTAP19−1 | 1556410_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6439)
KRTAP19−1 | 1556410_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.394)
KRTAP19−1 | 1556410_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRTAP19−1 | 1556410_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5446
p=0.651
p=0.2672
p=0.6865
p=0.8416
p=0.2293
p=0.059
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2